

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, Java, JavaScript, TypeScript, SQL, Bash

Data Pipelines & Distributed

Apache Spark / PySpark, Hadoop / MapReduce, distributed-data processing, high-throughput / low-latency tuning, ETL, SQL

ML Frameworks & Infra

PyTorch, scikit-learn, TensorFlow, NumPy, pandas, ML training-data pipelines, model evaluation, HuggingFace Transformers

Containers & Cloud

Docker, Kubernetes, AWS, GCP, Azure, Git, GitHub Actions, CI/CD, Linux

Distributed Systems

Scalability, reliability, fault tolerance, load balancing, multi-tenant systems, asynchronous messaging (AWS SQS), durable workflows

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Spark ETL Performance Optimization - Distributed Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL

- Profiled and optimized a **distributed Spark / PySpark ETL** + analytics pipeline over a **~1.4B-row** (~30GB) dataset, cutting end-to-end runtime **~25%** (17–19 min → 14.3 min) via broadcast joins, Adaptive Query Execution (AQE), shuffle-partition tuning, schema-aware column-pruned reads, and year/month partitioning with Snappy compression.
- Reduced analytics **tail-latency ratio (P99/P50) from 2.14x to 1.74x** through partition pruning, filter pushdown, pre-computed derived columns, and adaptive skew-join handling; localized shuffle/skew bottlenecks via Spark UI stage timings and documented a repeatable optimization playbook.

AgenticSearch - ML Pipeline with Provenance & Instrumentation

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite, Docker

- Built a multi-stage **ML/LLM pipeline** that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before model fallback, verifies outputs, and returns **cell-level provenance** for every result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, and pipeline instrumentation for inspecting extraction quality and ranking transparency.

Generative Models & Deep Learning - ML Frameworks from Scratch

github.com/Deva-1903/UMASS_CICS_689

PyTorch, JAX, NumPy | UMass CS 689 Advanced ML

- Trained and benchmarked **5 neural architectures** (perceptron, deep/ReLU MLP, VGG-style CNN, ResNet) on CIFAR-10 **across 3 optimizers** with learning-rate tuning, documenting train/test-error curves.
- Implemented generative models from scratch in JAX (a normalizing flow with coupling layers and a DDPM diffusion model) and built reverse-mode automatic differentiation over matrix-valued operations, learning training workflows from the ground up.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.